

topologies. Keong shares, "Multi-string designs have both merits as well as demerits. By having more rows of LEDs, the impact on the overall luminance is reduced in case an LED is damaged. However, uniform LED brightness in each row may not be achieved without the use of a multi-channel constant-current LED driver."

Multi-string LEDs are used in applications like automotive front-lights to keep the output voltage of the driver below a safety limit, and also in very high-power applications like street lamps and outdoor lamps. For these applications, Bianco says, "In general, topologies with multiple switching regulators (one per string) are better because of their better efficiency and lower power dissipation, while topologies with a single switching driver and multiple linear current sinks are not usable."

Currently, the use of multi-string solutions is very limited, probably because of their high cost. However, these will be used more in the future for red-green-blue lighting with controllable colours.

Thermal management

LED cooling is as always a challenge, especially for retrofit lamp applications—in particular, small form-factor ones (like MR16). LED is weak against heat and its lifespan decreases as the temperature increases. Therefore considering how to lower the heat is important when working with LEDs. Fortunately, multiple manufacturers are coming out with innovative solutions to improve cooling while keeping the weight of the heat-sink relatively low.

Thermal management is important in LED drivers as well as LEDs themselves. If the lamp is lit at a high temperature for long, its lifetime can reduce. Bianco suggests a solu-



ROHM LED driver suited for high-beam, low-beam and daytime-running light circuits

tion: "Electrolytic capacitors are in particular the most sensitive element to high temperature. So the design of an LED driver solution includes these capacitors. There are some drivers which work without electrolytic capacitors and whose performance is not much affected by a high temperature."

Other protections

An LED is basically a diode that is sensitive to current as well as EMI. In many applications like automotive, there is a significant EMI from other electronic and mechanical components. In such cases, reducing the EMI noise is essential. So drivers with EMI protection circuitry are used in these applications.

"Protection features such as over-voltage protection, short-circuit protection and under-voltage lock-out protection are important," shares Keong. "Another important feature of the LED driver IC is to have control over the safety aspects of an LED lighting application. This includes preventing the LED from damage and IC malfunction."

What's in store?

Higher efficiency, better uniform brightness, protection functions, lesser

component count and lower cost will be the keywords for LED drivers and solutions in the future.

Srinivasan says, "We foresee greater integration of LED power supply. There is also going to be greater integration of features like a single chip for AC/DC, multiple channels, PFC and dimming. New topologies like capacitor-less design (cap-less design) and single-stage high efficiency are also emerging."

Bianco feels, "For retrofit lamps, drivers will continue to improve to provide solutions that dim smoothly, while having low component count and solution cost. For other lighting

applications, there will be more drivers with complex features to control the light remotely."

Some vendors provide the ability to configure the driver in three different modes—step-down, step-up and auto-step (buck-boost)—by changing the external configuration. This, in turn, provides a high flexibility in application design.

Keong says, "With such flexibility, application designers can use the same IC to design different LED lighting applications. This will help them to minimise the inventory of multiple components."

The next big thing will be development of next-generation triac-dimmable LED driver ICs that draw on grid power. Semiconductor vendors will continue to support various reference designs requiring very few external devices. For example, no filter is needed to reduce the coupled EMI.

These new driver ICs will enable companies to design more compact LED driver boards at a lower cost. This, in turn, will make LED lighting systems more affordable and thus more widely used as light sources in homes and businesses. ●

The author is a senior technology journalist at EFY